 # Introduction to Numerical Weather Prediction

Chapter 3 : Objective Analysis And Data Assimilation

Objective

- ✓ *To Know The Process of Data Assimilation*
- ✓ *To Describe quality control mechanisms for Observed data*
- ✓ *To list Objective Analysis Schemes*
- ✓ *To be familiarize with Panofsky methods*

The Analysis Phase

- A gridded, 3-dimensional analysis is produced with
  1) A previous forecast
  2) Observations
- The process by which the above are combined is called **data assimilation**
- DA is the technique whereby observational data are combined with output from a numerical model to produce an optimal estimate of the evolving state of the system.

Data Assimilation

First guess contains background information retained by the model from previous analyses.

- Foot print of past weather conditions
- Uniform data coverage. Transports information from data rich regions to data void regions
- Parameters are physically consistent according to the governing equations.
- Model forecast has error.

Data Assimilation ..cont

Observations taken over a range of time, called a time window, usually centred on the analysis time

- Noisy
- Scattered, non-uniformly distributed
- Often insufficient to determine all the unknowns in the initial condition : the problem is *underdetermined*
- On many occasions indirectly related to the model variables

Data Assimilation ..cont

Observations are not perfect

- Instrumental Errors: Errors due to instruments not properly calibrated.
- Human Errors: Errors of human origin during computation or reporting the observation
- Representative Errors: Not representative of the grid averaged measurement of the parameter required by the model.

Data Assimilation ..cont

Quality control of Observations

- Bad reporting practice check: For human error in reporting the data
- Blacklist check: For stations which always report erroneous data
- Gross check: Checked against some reference e.g. climatology
- Background check: Based on the difference (called the "residual") between the observations and the expected value (which could be based on first guess)
- Buddy check: Comparing the observations with their "Buddies" or the nearby observations
- Analysis check: Quality control performed at the time of preparing analysis

Cont..

In words, the analysis equation can be expressed as shown below.

The analysis value at the grid point = The background [or first guess] value at the grid point + A correction

where

$$\text{correction} = \sum_{\text{sum}} \left(\text{weights} \times \text{new information [based on observations]} \right)$$

in other words

correction = a weighted average of a set of observations taken from sites surrounding the grid point

Data Assimilation

- Gridded atmospheric analyses are produced by combining the following:

  1) A previous forecast

  2) Forecast uncertainty

  3) Observations

  4) Observation uncertainty

Why we need data assimilation:

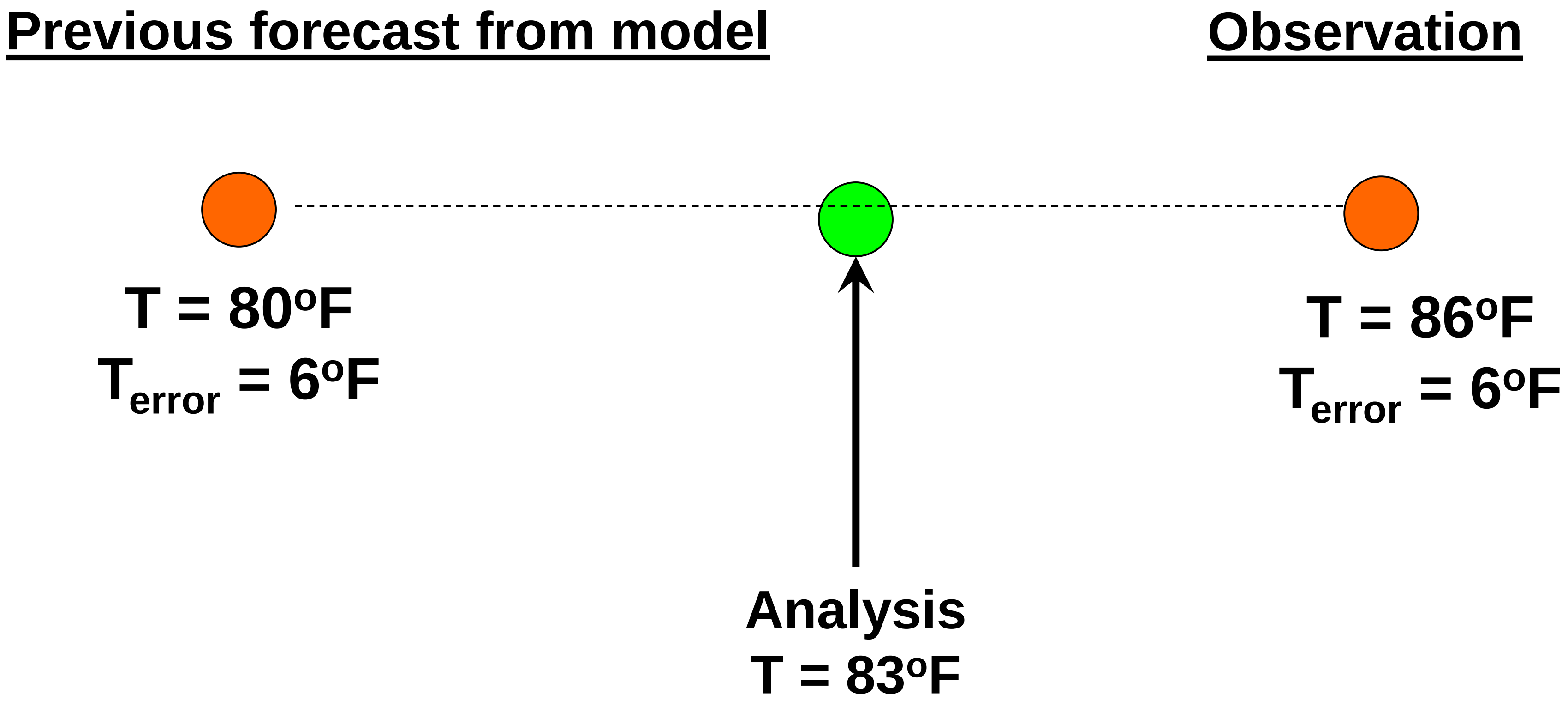
- **Data assimilation we required to understand**
  - range of observations
  - range of techniques
  - different errors
  - data gaps
  - quantities not measured
  - quantities linked

What are the benefits of data assimilation:

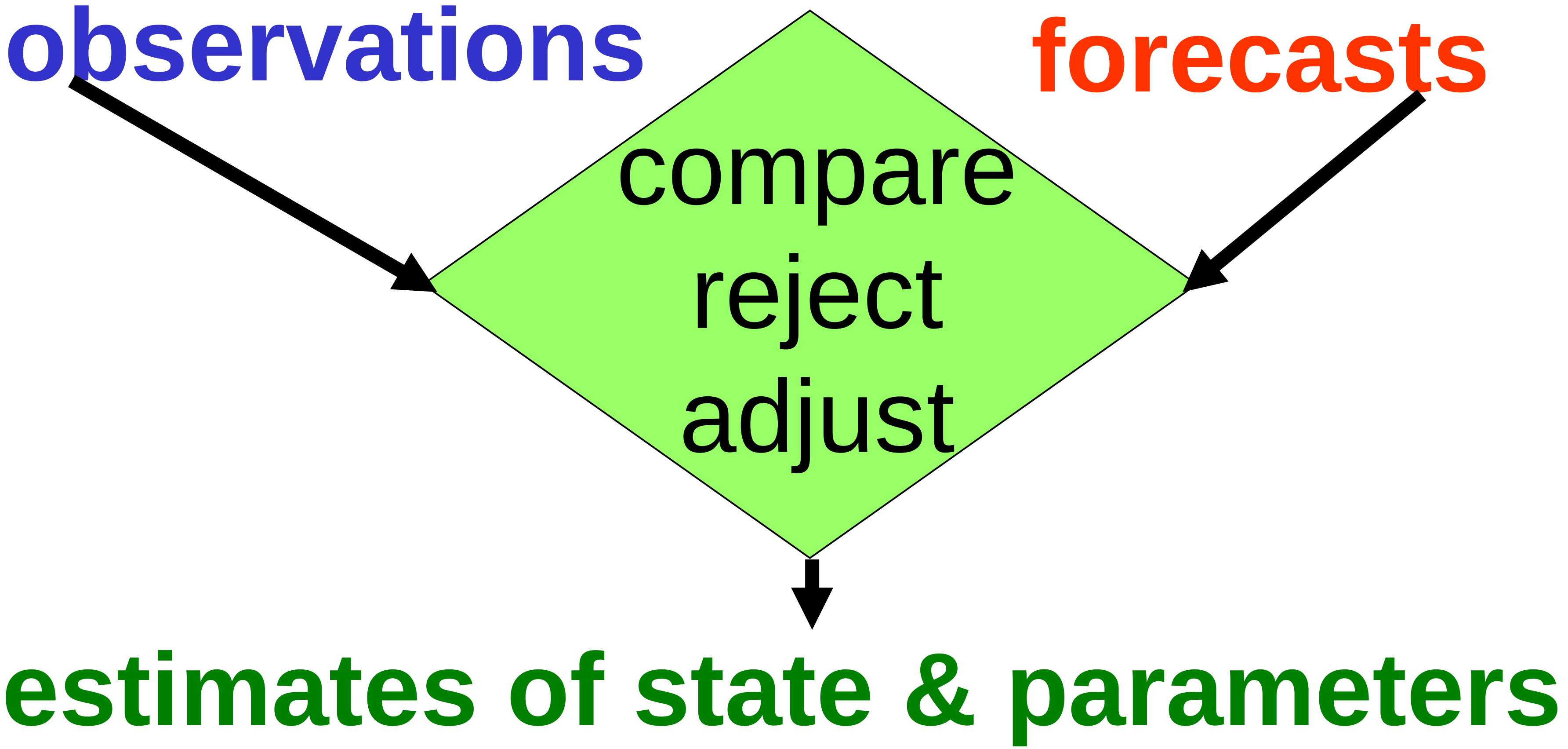
- Quality control
- Combination of data
- Errors in data and in model
- Filling in data poor regions
- Designing observational systems
- Maintaining consistency
- Estimating unobserved quantities

Data Assimilation

- Temperature at a single point

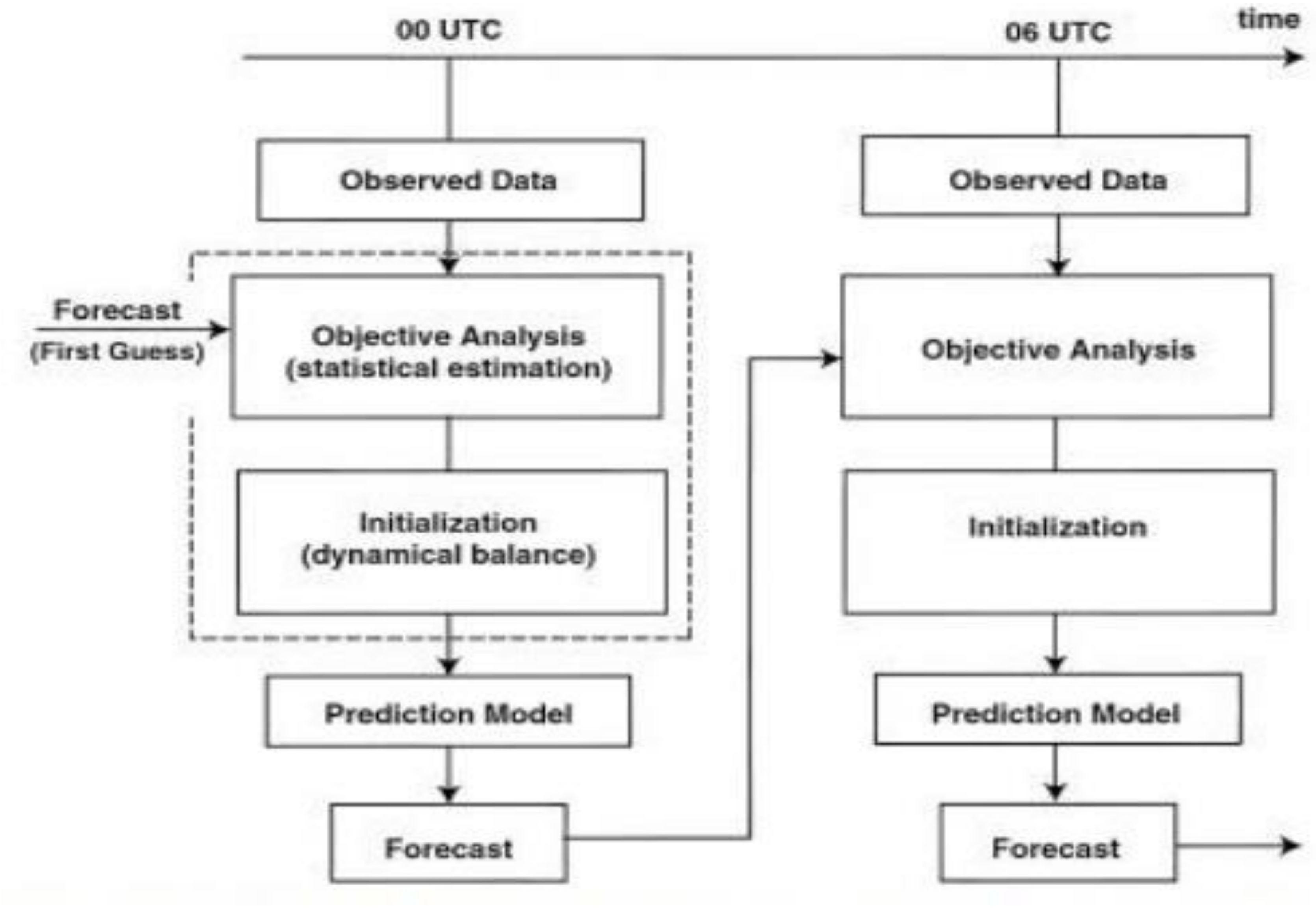


The Data Assimilation Process



errors in obs. & forecasts

Data Assimilation process

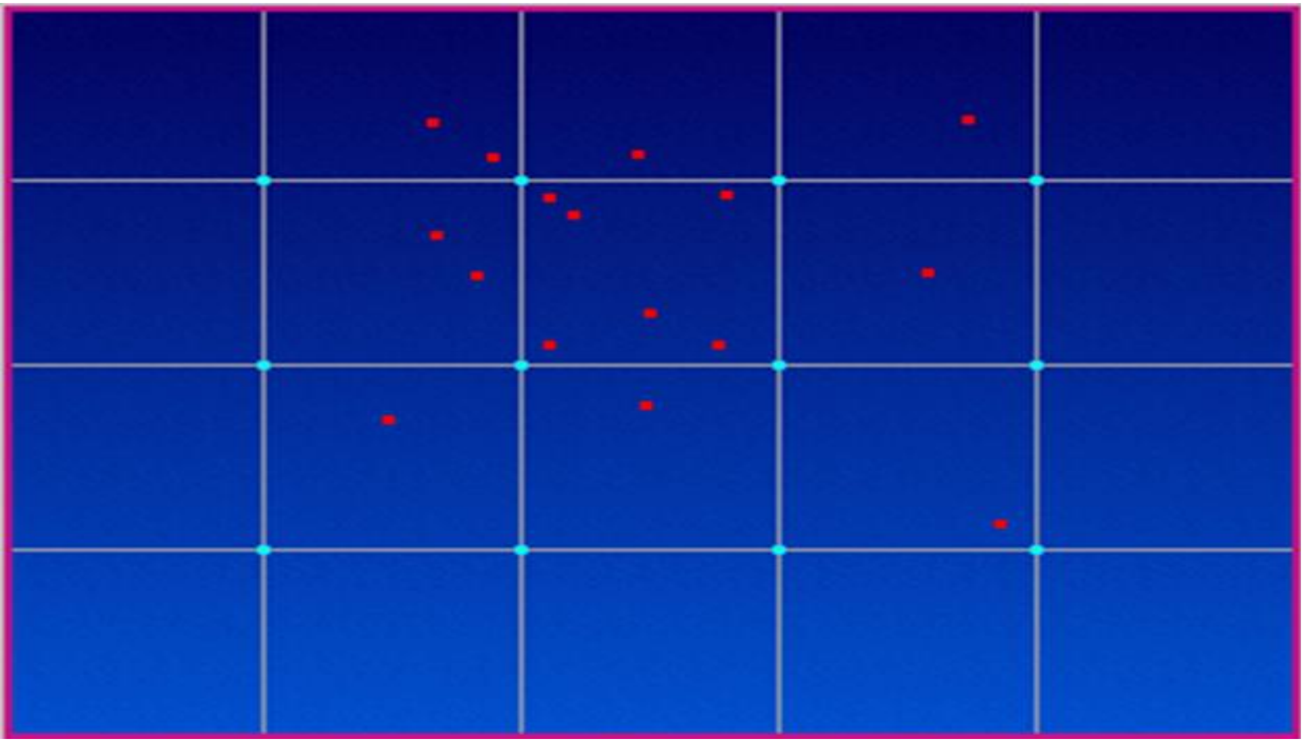


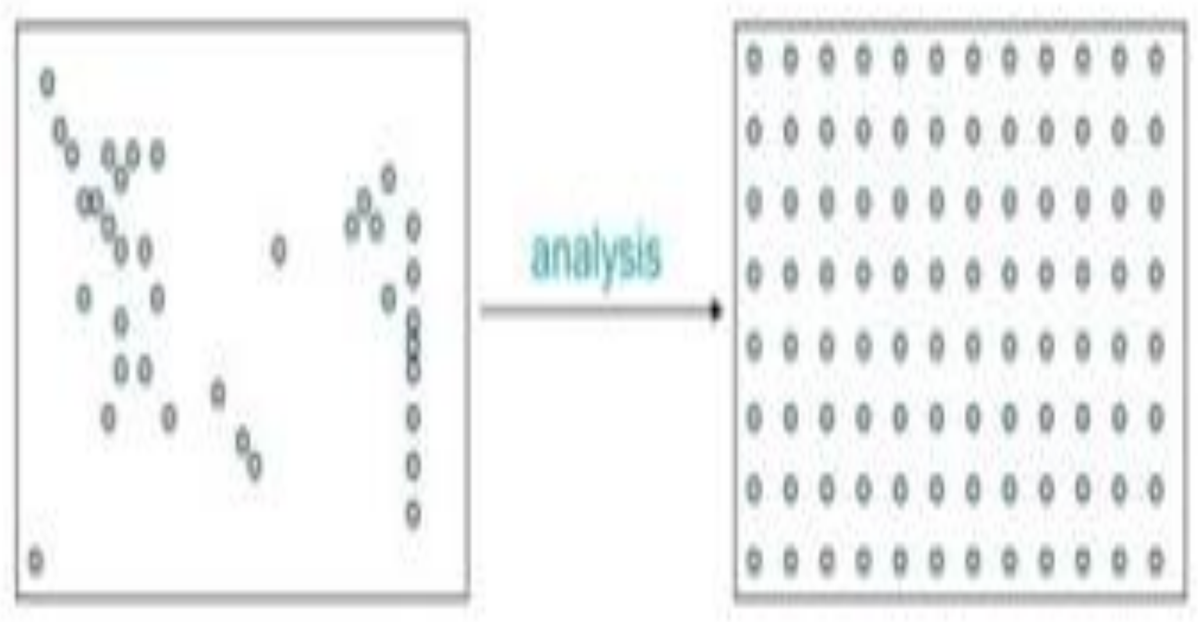
Objective analysis/data assimilation

- The goal is to produce an atmospheric state as close as possible to the reality and at the same time dynamically consistent, taking into account all the available information: observed data, model, physical constraints, climatology.

Data assimilation

- The process of interpolating observed values onto the grid points used by the model in order to define the initial conditions of the atmosphere is called **Objective analysis** .





Objective Analysis

Objective Analysis

- Numerical models require a set of initial conditions for the dependent variables (*u*, *v*, *w*, *T*, *q* etc.) at grid points which should come form synoptic or other data observed at stations.
- *Objective Analysis is the process which transforms information from randomly spaced observing stations into data at regularly spaced grid points.*
- An objective analysis scheme should perform a smooth interpolation, detect and remove bad data and carryout internal consistency analysis to reproduce the actual state of the atmosphere as accurately as possible.
- This reproduced state of the atmosphere is known as the *model state* or *analysis*.

Types of Objective Analysis

- Panofsky methods
- Cressman methods
- Barnes' methods

Function fitting method, Panofsky's or Polynomial Approach

- One of the first major contributions to objective analysis was suggested by Panofsky (1949).
- It consists of fitting a cubic polynomial to a given data set over a region.
- *It is the way of finding a polynomial function that passes through a given set of data point.*
- In this approach a set of observations is mathematically fit to a set of relatively smooth equations.

Objective Analysis

$p(x) = a_0 + a_1 x$	linear interpolation
$p(x) = a_0 + a_1 x + a_2 x^2$	quadratic interpolation
$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$	cubic interpolation

- In this method,the observations in area are fit to mathematical equation such as a polynomial equation(Panofsky,1949).

- Given the following n+1 data points

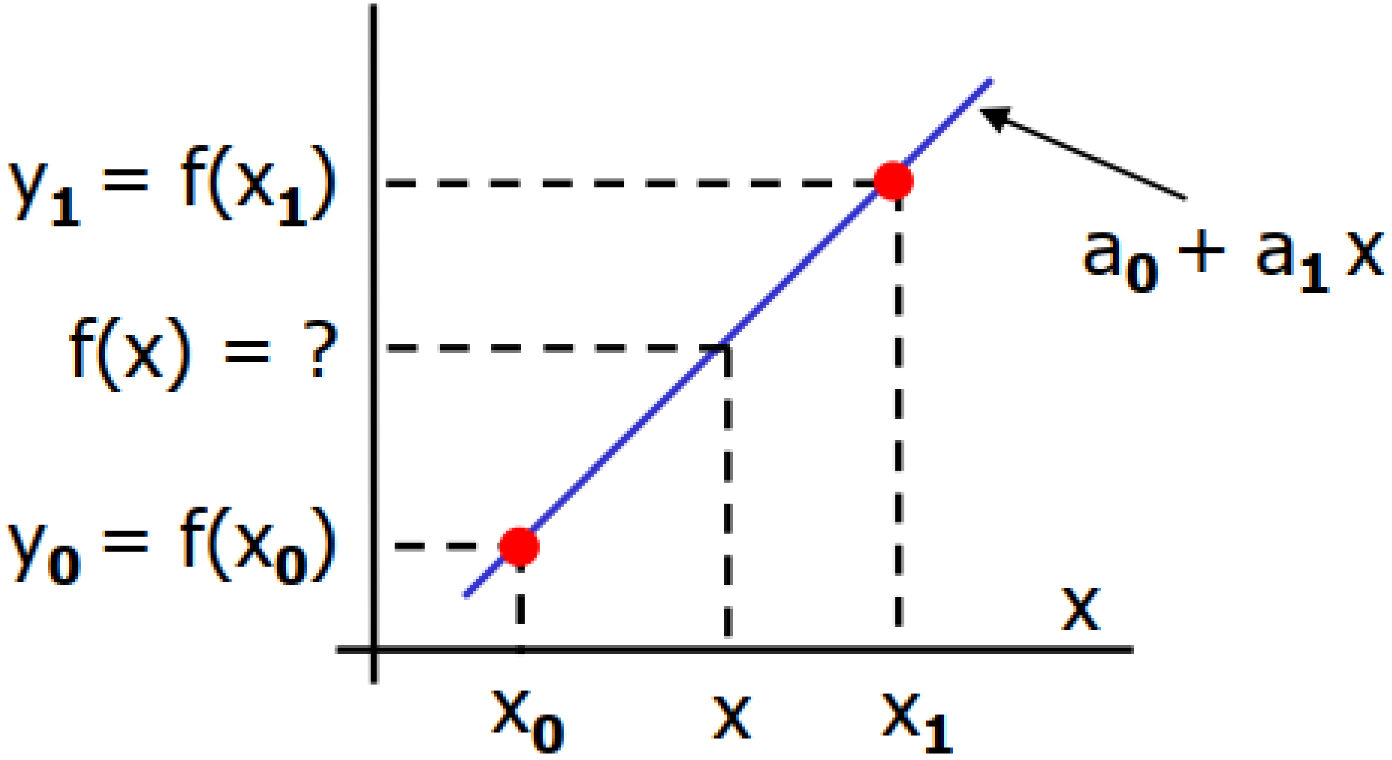
$$(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3), \quad \ldots, \quad (x_{n+1}, y_{n+1})$$

there is a unique n^{th} order polynomial that passes through them

$$p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$$

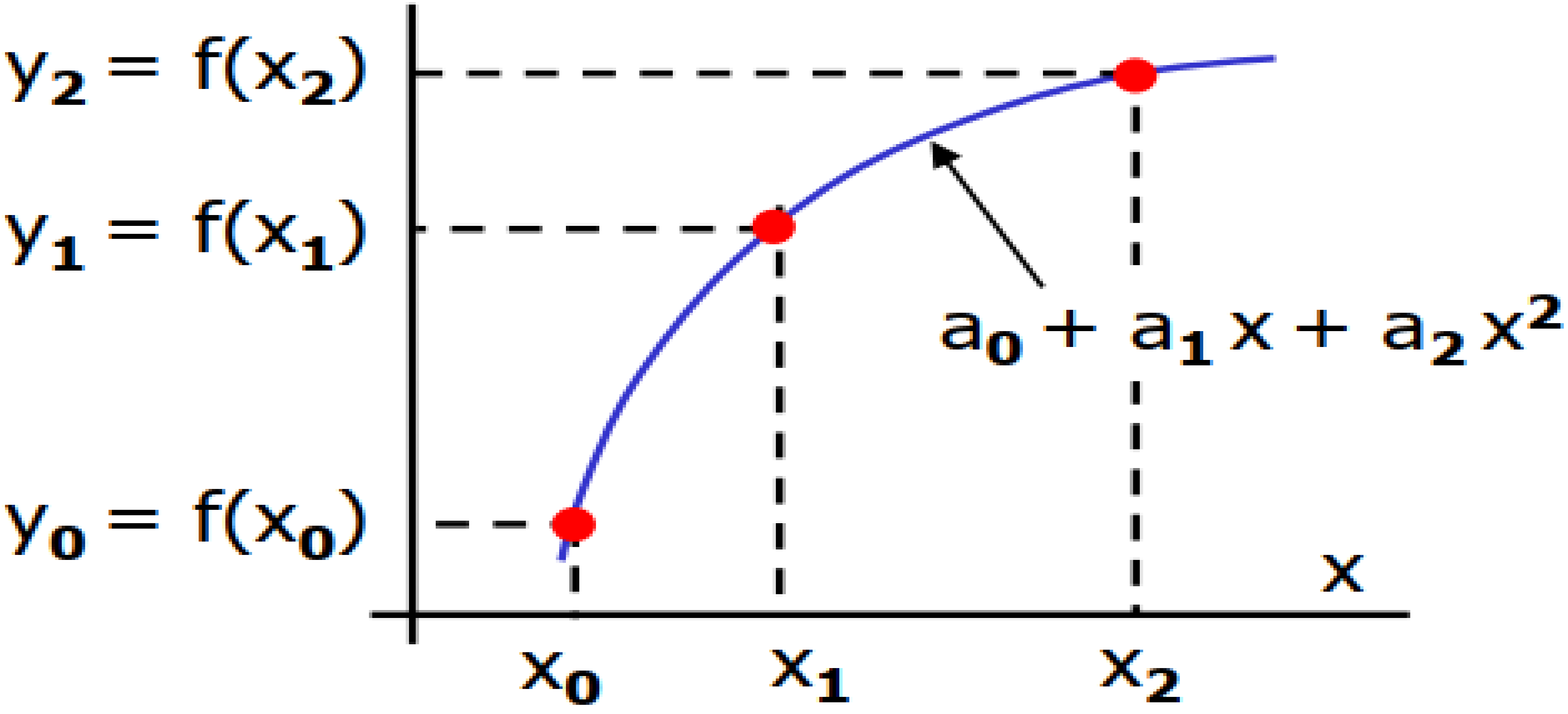
- The question is to find the coefficients $a_0, a_1, \ldots, a_n$

- **Linear Interpolation:**



Polynomial Interpolation

- **Quadratic Interpolation:**



Example 1

- Write the polynomial function for the given set of data

$f(0) = 1, f(2) = 2$

ANS

$$P(x) = 1+0.5x$$

Example 2

- Write the polynomial function for the given set of data

(1,1),(2,4),(4,5)

ANS

$$P(X) = -3.667+5.5X-0.833X^2$$

….END

Thanks !

 # Introduction to Numerical Weather Prediction

Chapter 3 : Objective Analysis And Data Assimilation

Objective

- ✓ *To Explain The steps involved in Cressman's method*
- ✓ *To be familiarize with Barnes' Objective Analysis Scheme*
- ✓ *To Describe The methods for Estimating vertical motion and stream function from wind field*
- ✓ *To Be aware of initialization of data on starting the model*

- ***Empirical Linear Interpolation***

Introduction: In this approach grid point values are determined as a linear combination of scalar variable values from nearby observing sites. The basic form of the linear equation is:

$$X_{grid} = \sum_{j=1 \text{ to } m} (h_j X_{oj})$$

where:

- *Xgrid = interpolated grid point value*
- *hj = scalar weight at point j*
- *Xoj = observation at point j*
- *m = number of observations*

- The scalar weight has the general form:

$$h_j = w_j \,/\, [\textstyle\sum_{j=1 \text{ to } m} (w_j)]$$

where:

- wj = weighting function applied to point j

❑ Note that the sum of all hj equals one [∑(hj)=1].

- The weight function (wj) can be determined in any number of ways.
- In general, empirically derived weight functions use distance weighting where observations that are closer to the grid point are given more weight than observations that are farther away.
- Ideally, all available observations can be used to determine the value at a grid point.
- If a weighting function falls off with distance, there is some distance away from the grid point where the weight becomes so small that the observations have a very little influence on the grid point value.

Radius of influence

- To save computer time and eliminate these low-influence observations, only observations within a specified distance from the grid point are used in the calculation. This distance is called the “radius of influence.”

Weighting Function Properties:

Weighting functions should have the following properties:

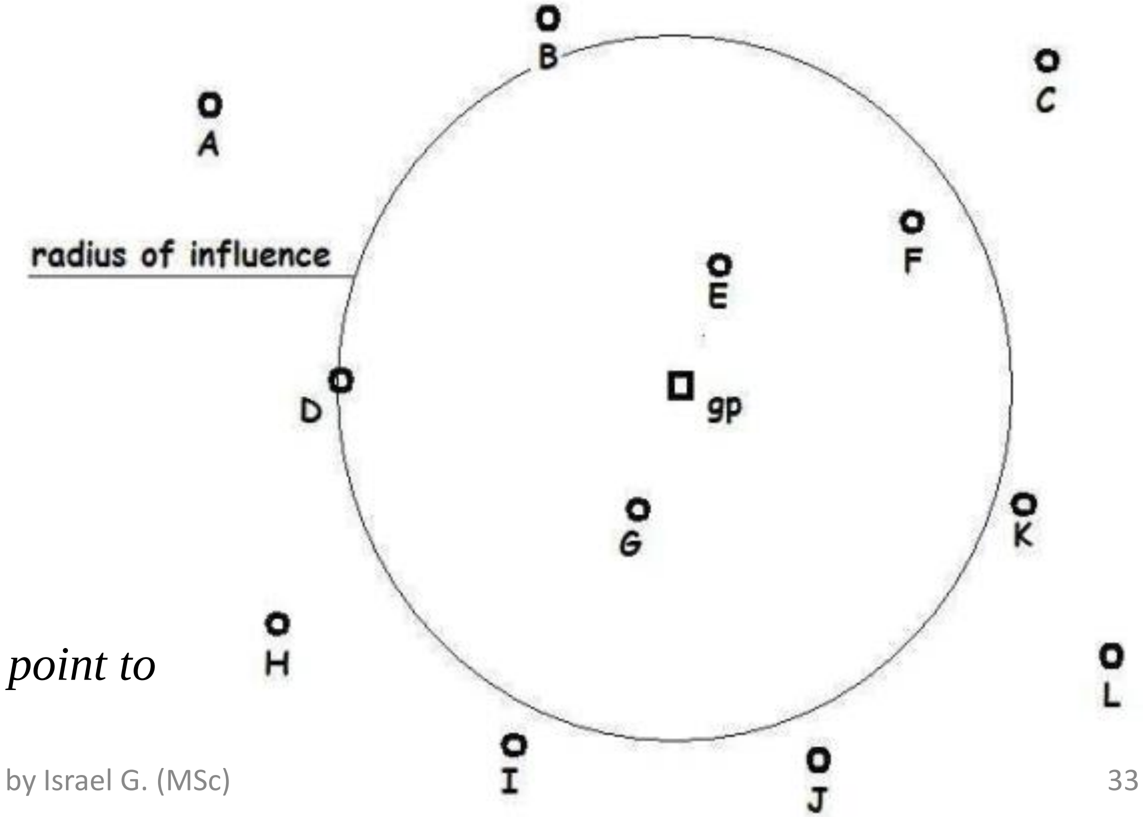
- ✓ The influence of an observation should be higher for closer observations and lower for observations farther from the grip point.
- ✓ If an observation is coincident with a grid point, it should have a unit weight.
- ✓The resulting grid values should be differentiable (a desirable property).
- ✓The more observations included in the interpolation, the higher is the smoothing level.

- ❑ Applying these properties result in a wide variety of weighting functions.

Example

- Let's consider the example shown in Figure below. There are 12 observations in the dataset. Only four observations are within the radius of influence: D, E, F and G. These will be used to calculate the value at the grid point (gp).

Point	X_{oj}=Ob	r
D	74	3.35
E	73	1.25
F	69	2.80
G	77	1.30



for r less than a 3.5 units, where r is the distance from the grid point to the observation (in arbitrary units).

ANS

The following table summarizes the calculations:

Point	X_{oj}=Ob	r	$w_j=e^{-r}$	h_j	h_j*X_j
D	74	3.35	0.0351	0.0536	3.97
E	73	1.25	0.2865	0.4375	31.94
F	69	2.80	0.0608	0.0928	6.40
G	77	1.30	0.2725	0.4161	32.04
Sum			0.6549	1.0000	**74.35**

Successive Corrections Method:

- The empirical linear interpolation uses the actual observation values when calculating grid point values.
- The successive corrections approach begins with a first guess or background grid and then uses the difference between these background values and the observations to determine the final grid point values via an iterative process.
- This method is useful for grid areas where observations are sparse or non-existent.

The successive corrections procedure includes a series of steps:

- Estimate the background value at each observation point [Xbj].
- Calculate the prediction error, that is, the difference between the background value at an observation point and the corresponding observation value [Xoj – Xbj].
- Use empirical linear interpolation to distribute the prediction errors to the grid points.
- Correct the grid point value based on the distributed errors.

[Xgrid(k+1) = Xgrid(k) + ∑j=1 to m (hj*{Xoj – Xbj})], where k is the iteration step index.

- Repeat the above steps until the value of all predicted errors is below a specified amount.

Cressman Scheme

- The Cressman (1959) scheme is described here because it is the first operational successive corrections OA scheme used by the Weather Bureau (now the National Weather Service) in the early days of operational numerical weather prediction.
- It is similar to the successive corrections scheme described in the previous section. It uses a non-zero weighting function within a prescribed radius of influence.

The Cressman weighting function takes the form:

$$w_j = (d^2 - r^2) / (d^2 + r^2) \qquad r < d$$

$$w_j = 0 \qquad r \geq d$$

where:

- r = distance from the grid point to the observation
- d = radius of influence

Barnes Scheme

- The Barnes (1964) proposed a scheme which could be used to interpolate randomly spaced data onto a grid with a desired level of detail. The Barnes weighting function is given by:

$$w_j = \exp\left[-r^2 / 4k \right]$$

- where:
  - r = distance from the grid point to the observation
  - k = parameter used to define the response of the weighting function

- The value of k is chosen so that when r=d (d is the radius of influence), the weighting function is e-4 times maximum value of data at r=0. The radius of influence is chosen so that it is greater than the average spacing of the observations.
- The iterative process recommended by Barnes for determining grid point values of some variable is as follows:

• On the first scan, estimate the value of the variable at each grid point using the following scheme:

$$x_{i,j} = [\ (\textstyle\sum_{n=1 \text{ to } N} w(r,d) * X_{on} \ ) \ / \ \textstyle\sum w(r,d) \ ]$$

where:

o xi,j = grid point value of the variable

o Xon = observed value of the variable

o N = number of observations

o w(r,d) = Barnes weighting function

- Next, estimate the variable value for each observation location by averaging the four closest grid values.
- Calculate the difference between the variable estimate and the observation value (residual).
- Distribute the differences using the linear interpolation scheme described above with the Barnes weighting function.
- Repeat the last three steps until the residual is less than a prescribed precision factor.

Calculation of vertical motion

Why Do We Care About Vertical Motion?

- We care about vertical motion for several reasons…
- Condensation, and thus cloud and precipitation formation, generally occurs from cooling an air parcel (or layer) to saturation through ascent.
- the mid-latitude cyclone and anticyclone formation, decay, and movement are functions of patterns of ascent and descent.
- Vertical vorticity amplification and generation are related to vertical motion and its horizontal gradients.

Calculation of vertical motion and stream function from wind field

- The vertical wind velocity is not an observed variable in meteorology, and is estimated from the observed horizontal wind field.
- The vertical velocity is an integral component of the three dimensional structure of the atmospheric motion and is encountered in many diagnostic and prognostic problems.

- The vertical motion can be estimated by using several methods:

a) Kinematic method

b) Adiabatic method

c)Vorticity method

d) Quasi-geostrophic omega equation

a) Kinematic method

- The simplest method for the computation of the vertical motion is the ***kinematic method***, in which the mass continuity equation would be integrated using the large scale horizontal wind observations, accounting for the divergence correction.

- The sparseness of observations constitutes a serious obstacle in this method. Moreover, due to uncertainties inherent to wind measurements, large errors are introduced in the calculation of the horizontal divergence and lead to important errors in the estimation of the vertical velocity.

b) Adiabatic method

- This method is based on the thermodynamic energy equation and is not very sensitive to errors in the measured wind field.
- In this method the temperature advection can be estimated quite accurately using the geostrophic wind, especially in mid-latitudes where observations are dense.
- This method can be employed when only geopotential and temperature data are available.
- Nevertheless, the adiabatic method involves the temperature tendency and is not recommended for a wide area unless observations are not too far apart in time.

Vorticity method

- The vertical velocity can also be estimated using a simplified form of the vorticity equation. In this method, the vertical advection of vorticity and the ‘twisting’ term are neglected and the relative vorticity is assumed to be small as compared to the Coriolis parameter in the divergence term.
- The time tendency and the horizontal advection of the geostrophic vorticity can then be estimated with a reasonable degree of accuracy.
- This leads to vertical velocity estimates more reliable than those obtained using the kinematic method.

d) Quasi-geostrophic omega equation

- This method is strictly diagnostic and estimates the vertical motion in terms of instantaneous values of geopotential.
- Furthermore, the use of the quasi-geostrophic omega equation does not require wind observations nor does it involve time tendencies, which makes it superior to all other methods.

Estimation of Stream function from wind

Estimation of Stream function from wind

- Wind derived quantities such as stream functions have introduced a great simplicity in the expression of the governing equations of the atmospheric motion and constitute the basic predictors for many numerical weather prediction models.

- It is, therefore, desirable to develop a transform pair which computes these potential functions given the observed wind field.

- The inverse transform would provide the wind field from the stream functions.

Computation of Stream Function

For nondivergent horizontal motion the flow field can be represented by a *Stream function* Ψ (x,y) so that,

$$u = -\frac{\partial \psi}{\partial y}$$

$$v = \frac{\partial \psi}{\partial x},$$

The vorticity is then given by

$$\zeta = \partial v/\partial x - \partial u/\partial y = \partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 \equiv \nabla^2\psi \quad \text{(Poisson equation)}$$

From the vorticity field stream function can be computed by inverting the Laplacian using a numerical method (iterative methods, i.e., Relaxation method, or Jacobi's or Gauss-Seidel methods).

Estimation of Stream function from wind

- The equations of the above type are referred to as *elliptic* partial differential equations. Since it has no time dependency term, we are not using it to predict the stream function at a future time.
- We are only using it to *diagnose* the stream function from other known values at the current time. Elliptic equations are difficult to solve numerically.
- There are several techniques that can be used to solve the equations e.g *Relaxation Method and Fourier Transform Method.*

❖ ***Reading Assignment***

- *Relaxation Method*
- *Fourier Transform Method*
- *Geopotential Height from Wind Field*

Model Initialization

- The process by which observations are *processed* to define the initial conditions for a model's dependent variables.
- (where *processed* infers taking observations, performing quality control, assimilating the observations, and ensuring that the final analysis is balanced and representative of the real atmosphere).
- The process of entering observation data into the model to generate initial conditions is called *initialization*.

Initialization & filtering

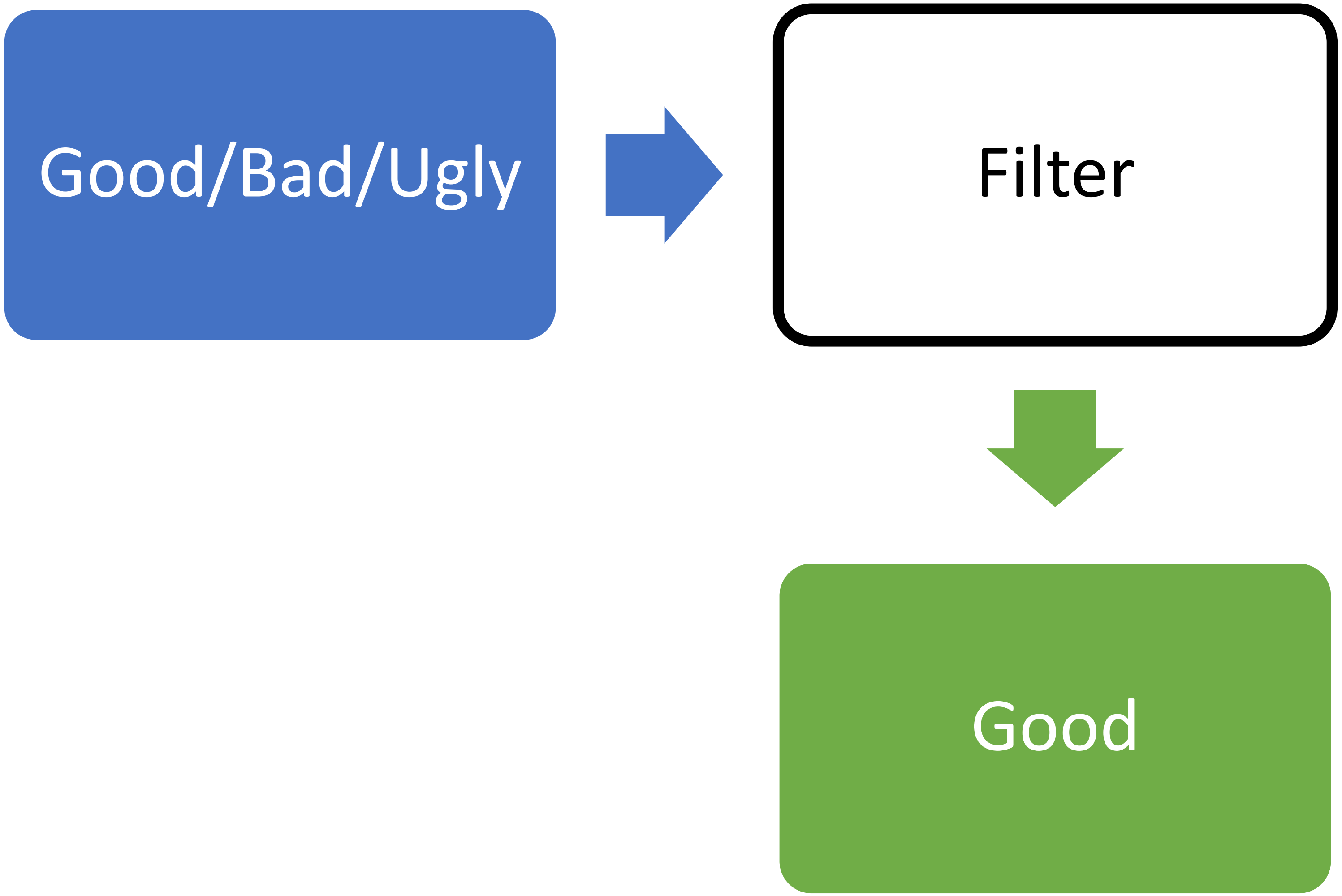
- The process of providing initial value data to a model is known as *initialization*.
- Initialization strategy depends on the scale of the phenomena that we want to forecast
- Models for synoptic scale prediction are usually initialized every 6 hours using different types of observations (soundings, satellite, surface, etc)

Initialization and filtering

- Models for mesoscale forecasts have to be initialized more frequently (1 hour to 15 minutes) using dense observational networks , radars and other observations when they are available.
- The **smaller** the scale the **larger** the number of observations that we need and the higher the assimilation frequency.

Initialization &Filtering

- Extracts the signal from the noisy observations (**filtering**)



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Thanks !